Warmup/Attendance: [Week 6: Wednesday 9/30/2026]

Full Name: SID:

Work with the people around you to try to solve these problems, which we'll discuss together once lecture begins. These are not graded for correctness, but you will turn this sheet in for attendance at the end of class.

For the following set of questions, you'll be presented with a scenario and then asked to identify whether the scenario describes a

1. **Binomial** distribution (if so, what are n and p?)
2. **Hypergeometric** distribution (if so , what are N, G, and n?)
3. *Neither* of these (why)?

1. **Binomial, hypergeometric, or neither?**

Roll a fair ten-sided die 20 times. We count the number of times we roll a multiple of 3.

2. Binomial, hypergeometric, or neither?

YouGov surveyed 1,500 adult US citizens in December 2023, and counted the number of respondents who had read at least one book that year. The population of the US is about 335 million.

3. Binomial, hypergeometric, or neither?

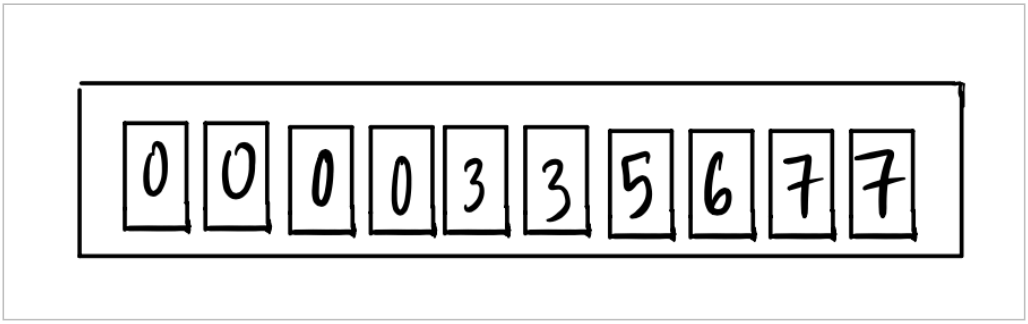
A six-sided die is tossed twice, and we check if the sum of the spots is 8 or not.

4. Binomial, hypergeometric, or neither?

A bag that has 6 pieces of fruit: 2 mangoes, 3 apples, and 1 orange. I reach into the bag and draw out one fruit at a time, selecting each fruit at random (so all the fruit left in the bag are equally likely on each draw). I put the fruit away after I take it out (and don't put it back into the bag). I count the number of draws until and including the first time I draw a apple.

5. Match the histogram to its description

Consider the following box of tickets. Match each of the following (lettered) plot to its accurate description as given on page 2 your warm up sheet:



Match the description number below to the letter of the plot on the slide:

1. The probability histogram for the value of a ticket drawn at random from the box
2. An empirical histogram for which the data were generated by drawing **10** tickets from the box with replacement
3. An empirical histogram for which the data were generated by drawing **100** tickets from the box with replacement
4. An empirical histogram generated by **20** draws from a different box.